MIL-W-12404E(EL)
7 July 1970

Superseding
MIL-W-12404D(EL)
30 January 1968

MILITARY SPECIFICATION
WIND MEASURING SET AN/GMQ-11( )
(Units of)

1. SCOPE

1.1 This specification covers the following units of one type of wind measuring set, designated Wind Measuring Set AN/GMQ-11( ). (See 6.2)

Transmitter, Wind Direction and Speed T-420( )/GMQ-11
Indicator, Wind Direction and Speed ID-373( )/GMQ-11
Support, Wind Direction and Speed Transmitter MT-1246( )/GMQ-11

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of the invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, wrapping
QQ-S-781	Steel strapping, Flat
PPP-B-20	Bags, cotton, mailing
PPP-B-585	Box, wood, wirebound
PPP-B-601	Boxes, wood, cleated-plywood
PPP-B-621	Box, wood, nailed and lock-corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, corrugated and solid, sheet stock (container grade), and cut shapes
PPP-P-291	Paperboard, wrapping, cushioning
PPP-T-45	Tape, gummed, paper, reinforced
PPP-T-76	Tape, pressure-sensitive adhesive paper, water resistant, (for carton sealing)

FSC-6660

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MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials, and Processes used in Signal Corps Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-C-45662	Calibration of Standards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-810	Environmental Test Method for Aerospace and Ground Equipment

DRAWINGS

ELECTRONICS COMMAND

SC-DL-279855	Transmitter, Wind Direction and Speed T-420( )/GMQ-11
SC-DL-279856	Indicator, Wind Direction and Speed ID-373( )/GMQ-11
SC-DL-279857	Support, Wind Direction and Speed Transmitter MT-1246( )/GMQ-11

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - Wind Measuring Set AN/GMQ-11 is a fixed unit consisting of a Transmitter, T-420, Indicator, ID-373 and Transmitter support, MT-1246 that gives visual indications of wind speed and direction. The propeller driven generator produces voltage linear to wind speed and a synchro system is used for measuring wind direction.

3.2 Preproduction samples. - The contractor shall furnish preproduction samples for approval as required by the invitation for bids and contract. (See 6.1(c)(1)). When preproduction samples are not specified by contract or purchase order four (4) production units of the first ten (10) produced shall be tested to the same requirements as specified for preproduction samples under 4.3. (See 4.8.2.1)

3.3 Construction. - The equipment shall be constructed in accordance with the requirements of this specification and equipment drawings listed in 2.1. The requirements of MIL-P-11268 shall apply when the part, material or process is not specified by drawing.

3.4 Finish. - The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4)

3.5 Marking. -

3.5.1 General. - Marking shall conform to MIL-M-13231 and the equipment drawings. (See 4.4)

3.5.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. - All items listed in 1.1 shall be serial numbered, except Support, Wind Direction and Speed Transmitter MT-1246( )/GMQ-11.

3.6 Tropicalization of material. - In addition to the requirements of MIL-P-11268 and the drawings the following is required:

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3.6.1 Fungus. - Equipment and assemblies shall show no evidence of viable fungus or corrosion* when subjected to 28 days of fungus test in accordance with paragraph 4.9.5.

3.6.2 Salt Fog. - The equipment and assemblies shall be exposed, in their operating configuration, to 48 hours of salt fog in accordance with paragraph 4.9.6; and shall show no evidence of corrosion.

3.7 Performance. -

3.7.1 Indicator ID-373( )/GMQ-11. -

3.7.1.1 Wind speed accuracy. - The indicator shall conform to the test of 4.8.1.1. Readings shall be within the prescribed limits for both knots and miles per hour. (The meter resistance shall be 2300 ± 11.5 ohms with the meter accuracy within ± 1 percent.)

3.7.1.2 Direction accuracy. - The direction indicator shall indicate a position within ± 2 degrees of a transmitted simulated wind direction signal as derived from a standard, calibrated, synchro transmitter.

3.7.2 Transmitter T-420( )/GMQ-11. -

3.7.2.1 Voltage output vs wind speed. - The transmitter wind speed output voltage shall be linear with respect to wind speed and shall be 0.1216 volt per knot with a total tolerance of ± 0.27 volt at wind speeds of 3 to 60 knots; ± 0.48 volt at wind speeds of 61 to 120 knots; and ± 0.65 volt at wind speeds of 121 to 160 knots.

3.7.2.2 Output voltage vs shaft speed. - The output voltage vs shaft speed shall conform to 4.8.2.2.

3.7.2.3 Starting and stopping speeds. - The transmitter shall have a stopping speed of not more than 2.4 knots, and a starting speed less than 3.4 knots.

3.7.2.4 Torque, starting. - The shaft shall start and continue to rotate when a constant torque of 0.70 inch-ounces is applied in the counter-clockwise direction.

3.7.2.5 Torque, stopping. - The shaft rotation shall stop at a torque load of less than 0.45 inch-ounces.

* Corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

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3.7.2.6 Vane balance. - The vane balance shall conform to the test of 4.8.2.5.

3.7.2.7 Vane accuracy. - The vane shall align the transmitter to within three (3) degrees of the true wind direction.

3.7.2.8 Vane shaft bearing friction. - The vane shaft bearing friction shall conform to the test of 4.8.2.6.

3.7.2.9 Synchro slippage. - All synchros shall conform to the test of 4.8.2.7.

3.7.3 Electromagnetic compatibility (RFI). - The Wind Speed Transmitter T-420( )/GMQ-11 shall meet the limits of MIL-STD-461, Class ID. (See 4.4). Application of suppression components and techniques shall conform to that specification. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance with the specified emanation and susceptibility limits, he shall obtain approval from the contracting officer before proceeding with the modification. MIL-STD-461 is not required for Weston Model 750 Generator, series A14R or A16R.

3.7.4 Overall operation. - The transmitter and indicator shall be connected as in normal operation. The propeller shall be removed and the magneto shaft rotated at the specified speeds in 4.8.2.2, the indicated speeds shall meet the limits in knots of 4.8.1. The wind direction shall be as specified in 4.8.1.2.

3.8 Service conditions. - The equipment shall meet the following service conditions where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.8.1 Operation. - Continuous use for a period of one (1) year, under the world-wide environmental conditions specified in 3.8.2 through 3.8.5 with a continuous duty cycle and no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement.

3.8.2 Temperature (See 4.9.1). -

(a) Operating: Ambient temperature in the range of +150°F to -40°F. (The 150°F temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

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(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.8.3 Relative humidity (See 4.9.2). - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation for 4 hours.

3.8.4 Rain (See 4.9.3). - Two inches of rainfall (fresh water) per hour, with wind at 20 miles per hour.

3.8.5 Shock; bench handling. - (Indicator ID-373( )/GMD-11 only). - The Indicator shall meet full specification requirements after subjection to the shock tests of 4.9.4.

3.9 Preconditioning. - Indicator ID-373( )/GMQ-11 shall be capable of meeting the requirements herein, without a subsequent processing, after subjection to the bounce preconditioning of 4.6. (See 4.5)

3.10 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.10). Individual items shall not be hand-picked for fit or performances. Reliance shall not be placed on any unspecified dimensions, rating, characteristics, etc.

3.11 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the following applicable portions of MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
  Slack.
  Protection.
  Clearance
  Splicing.
  Connections, general
  Grounding, general
  Shielding on wire and cable
Soldering and brazing.
Cleaning of units.
General requirements for securing of parts.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.
Cleaning prior to welding.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless otherwise disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.11)

4.3 Preproduction inspection. - This inspection will be performed by the contractor unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (See Tables I, II and III, respectively) and performed in that order except that Overall Operation Table I shall be last.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Finish	3.4
Marking	3.5
Electromagnetic Compatibility	3.7.3

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4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the government at its discretion may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6). Test equipment for government verification inspection shall be made available by the contractor. The indicator ID-373( )/GMQ-11 shall be preconditioned after final assembly.

4.5.1 Group A inspection. - This inspection, including sampling, shall conform to Table I and the ordinary inspection procedures of MIL-STD-105 using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the government except Overall Operation shall be last.

TABLE I - GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.11	4.7		
Support ID-373( )/GMQ-11	3.11	4.7	1.5 dphu*	6.5 dphu*
Transmitter T-420( )/GMQ-11	3.11	4.7	1.5 dphu	6.5 dphu
Indicator ID-373( )/GMQ-11	3.11	4.7	1.5 dphu	6.5 dphu
Performance	3.7	4.8		
Indicator ID-373( )/GMQ-11	3.7.1	4.8.1		
Wind speed accuracy	3.7.1.1	4.8.1.1	1.0% For The Group	All Electrical Defects Are Major
Direction accuracy	3.7.1.2	4.8.1.2		
Transmitter T-420( )/GMQ-11	3.7.2	4.8.2		
Output voltage vs shaft speed	3.7.2.2	4.8.2.2		
Starting torque	3.7.2.4	4.8.2.3		
Stopping torque	3.7.2.5	4.8.2.4	1.0% For The Group	4.0%
Vane shaft bearing friction	3.7.2.8	4.8.2.6		
Overall operation	3.7.4	4.8.2.8	1.0%	

*dphu: defects per hundred units.

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4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection.

4.5.2.2 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the government.

Table II - Group B inspection

Inspection	Requirement Paragraph	Inspection Paragraph
Interchangeability	3.10	4.10
Vane balance	3.7.2.6	4.8.2.5
Synchro slippage	3.7.2.9	4.8.2.7

4.5.3 Group C inspection. - This inspection shall consist of the tests specified in Table III and shall be performed on sample units that have been subjected to and met Group A and B inspections. Sample units shall be selected in accordance with paragraph 4.5.3.1.

Table III - Group C Inspection

Inspection	Requirement Paragraph	Inspection Paragraph
Sub-Group I		
Temperature	3.8.2	4.9.1
Moisture-resistance	3.8.3	4.9.2
Rain	3.8.4	4.9.3

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Table III - Group C Inspection (contd)

Inspection	Requirement Paragraph	Inspection Paragraph
Sub-Group II		
Shock test (bench handling)	3.8.5	4.9.4
Voltage output vs wind speed	3.7.2.1	4.8.2.1a
Starting speed	3.7.2.3	4.8.2.1b
Stopping speed	3.7.2.3	4.8.2.1c
Vane accuracy	3.7.2.7	4.8.2.1d
Fungus	3.6.1	4.9.5
Salt fog	3.6.2	4.9.6

4.5.3.1 Sampling for inspection of equipment. - Four items of production shall be selected for Group C inspection from the first 10 items produced and subjected to Sub-Group I & II inspection. One item shall be selected from each 20 items produced thereafter shall be subjected to the inspection of Sub-Group I. The items shall be selected without regard to their quality and shall be selected at random from each group of units produced. When preproduction samples (First Articles) are specified by contract or purchase order Group C inspection is not required for the four production items of the first 10 produced.

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the process and materials, as applicable, and, all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

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4.5.4 Reinspection of Conforming Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A and Group C inspection.

4.6 Bounce preconditioning. - Indicator ID-373( )/GMQ-11 shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shallhave a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 3 minutes. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Visual and mechanical inspection. - The equipment shall be examined for the defects listed in MIL-STD-454 and Table IV.

TABLE IV - CLASSIFICATION OF VISUAL AND MECHANICAL DEFECTS

Classification	Defect
Major	Transmitter does not rotate freely or evenly. Propellor does not retate freely or evenly. Indicating arrows do not move freely or evenly. Support bent. Tripod legs bent or not of equal lengths. Impeller vanes bent. Tail vane bent or dented. Terminal board broken. Synchro mounting inadequate.
Minor	Tripod mounting foot bent. Tripod mounting foot missing. Light control difficult to operate or loose.

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4.8 Performance. - Test equipment shall conform to MIL-C-45662 and the nominal input power of the Indicator ID-373( )/GMQ-11 is 115 VAC, 60 Hz. Before making any performance tests on the transmitter the direction synchro shall be electrically zeroed.

4.8.1 Indicator ID-373( )/GMQ-11. -

4.8.1.1 Wind speed accuracy. - Apply the following d.c. voltages across the terminals of the speed indicator and the readings shall indicate speeds within the listed limits.

Applied voltage	Speed indication in Knots
14.60	120 ± 1.2
9.73	80 ± 1.2
6.08	50 ± 1.2
2.43	20 ± 1.2
1.22	10 ± 1.2
0.61	5 ± 1.2

Applied voltage	Speed indication in mph
12.67	120 ± 1.2
8.45	80 ± 1.2
5.28	50 ± 1.2
2.11	20 ± 1.2
1.05	10 ± 1.2
0.53	5 ± 1.2

4.8.1.2 Direction accuracy. - With the transmitter and indicator connected as in normal operation energize the 115 volt circuit. Set the vane to the zero position (North), note the corresponding reading on the direction indicator dial. Rotate the vane clockwise stopping at 30° point and noting the dial reading. Repeat this test in the counter-clockwise direction. The noted readings shall conform to 3.7.1.2.

4.8.2 Transmitter T-420( )/GMQ-11. -

4.8.2.1 Preproduction or first production transmitter tests (See 3.2 & 4.5.3.1). - The following tests shall be performed by placing the complete transmitter mounted as in normal operation in a wind tunnel at the National Bureau of Standards, Washington, D. C.

(a) Voltage output vs wind speed. - The measured output voltage shall conform to 3.7.2.1 when the unit is subjected to the following speeds; 5, 10, 20, 30, 40, 50, 60, 80, 100, 120, 140 and 160 knots. The actual wind speeds used shall be within ±5 knots or ±10%, whichever is lesser of the specified speed.

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(b) Starting speed. - Increase the wind speed gradually until the propeller starts, to determine compliance with 3.7.2.3. At least five tests shall be made and readings taken of which the average shall be the starting speed.

(c) Stopping speed. - Decrease the wind speed gradually until the propeller stops, to determine compliance with 3.7.2.3. At least five tests shall be made and readings taken of which the average shall be the stopping speed.

(d) Vane accuracy. - Position the equipment in a 4.4 knot wind and record the wind direction indication. With the wind continuing manually displace the wind vane eight (8) degrees, first in one direction then in the other, and then allow it to find its own center in the 4.4 knot wind stream. The wind direction indication shall conform to 3.7.2.7.

4.8.2.2 Output voltage vs shaft speed. - The propeller shall be removed from the transmitter and replaced with a rotating device whose speed can be controlled within ±1 rpm. A 1150 ±5.75 ohm load (including voltmeter) shall be connected across the magneto output terminals. The voltage output and the magneto shaft speed shall be within the limits specified for Table V.

TABLE V

Magneto shaft rpm	Voltage output
2856 ± 29	14.6
1904 ± 29	9.73
1190 ± 29	6.06
476 ± 29	2.43
238 ± 29	1.22
119 ± 29	0.61

4.8.2.3 Torque test, starting. - Apply a constart torque of 0.70 inch-ounces to the impeller shaft in a counter-clockwise direction (viewed from the impeller end or front) to determine compliance with 3.7.2.4.

4.8.2.4 Torque test, stopping. - Start the shaft in a counter-clockwise direction by applying a torque of 0.75 inch-ounce and measure the torque of the free turning shaft to determine compliance with 3.7.2.5. A torque of 0.45 inch-ounce or greater does not pass the test.

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4.8.2.5 Vane balance. - Mount the transmitter in a horizontal position in such a manner that the vane portion is free to rotate. Rotate the vane so that its long axis is horizontal and observe the position assumed by the vane when free to rotate. The vane is considered balanced when not more than 5 ounce-inches of torque applied to the lighter end will restore the vane to a horizontal position.

4.8.2.6 Vane shaft bearing friction. - With the transmitter mounted as in normal operation, apply 1.75 ounce-inches of torque to the transmitter rain shield. The vane shall continue to rotate as long as the torque is applied.

4.8.2.7 Synchro slippage. - With the synchro mounted as in normal operation, apply a torque of at least two (2) pound-inches to the casing. The synchro shall not rotate while subjected to this test. Apply the same torque in the opposite direction and test.

4.8.2.8 Overall performance. - The equipment shall conform to 3.7.4.

4.9 Service condition tests. -

4.9.1 Temperature test (Transmitter and Indicator only). - The equipment shall be subjected to the temperature cycle shown in MIL-STD-169. Performance shall be as follows:

Step	Paragraph	Measurement	Performance
1,3, 8 and 10	4.8.1.2	Direction accuracy	Full specification requirements
	4.8.1.1	Wind speed accuracy	
	4.8.2.2	Output voltage vs Shaft speed	
	4.8.2.5	Vane balance	
	4.8.2.3	Starting Torque	
	4.8.2.4	Stopping Torque	

4.9.2 Moisture-resistance test (Transmitter and Indicator only). - The equipment shall be tested as follows:

4.9.2.1 Test conditions. -

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

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4.9.2.2 Test method. -

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.9.2.3, and readjust or re-align as necessary to meet full specification requirements.

(d) The equipment shall be subjected to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. The measurements specified in 4.9.2.3 shall conform to MIL-STD-170. The measurements specified in 4.9.2.3 shall be made during the periods shown on the standard and shall comply with 4.9.2.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.9.2.3.

4.9.2.3 Performance. - Measurements made during this test and the specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
4.8.1.2	Direction accuracy	All specification
4.8.1.1	Wind speed accuracy	requirements
4.8.2.5	Vane balance	
4.8.2.3	Starting torque	
4.8.2.4	Stopping torque	
4.8.2.6	Vane shaft bearing friction	
4.8.2.2	Output voltage vs shaft speed	

4.9.2.4 Failure. - If any equipment fails to meet the performance specified in 4.9.2.3 during cycling, it shall be re-aligned or readjusted once. If the equipment then fails to meet 4.9.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.9.2.2(e), it does not pass the test.

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4.9.3 Rain test (Transmitter only). - The equipment shall be tested as follows: (NOTE: During this test the transmitter shall be restrained from rotating on its shaft).

(a) Dry at 150° ±5°F for 48 hours.

(b) Condition for 4 hours at 77° ±5°F and 40 to 60 percent relative humidity.

(c) Take the measurements specified in 4.8.2.8 to insure that the equipment meets specified performance.

(d) Place the equipment, in its normal operating position, in a test chamber maintained at 77° ±5°F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ±5°F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

(e) With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ±5 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for one hour.

(f) Rotate the support 90 degrees horizontally, to expose a second side of the component to the wind and rain. Maintain this condition for one hour.

(g) Continue as in 4.9.3(f) until all four sides of the support has been exposed to the wind and rain, each side for one hour.

(h) Immediately after completion of the foregoing, repeat the measurements specified in 4.9.3(c) without removal from the chamber but with the rain and wind discontinued. Complete the measurements within one hour. The equipment shall meet specified performance.

(i) Open Transmitter T-420( )/GMQ-11. There shall be no visible evidence of leakage.

4.9.4 Shock test: Bench-handling. - Indicator ID-373( )/GMQ-11 shall be placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

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(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.9.5 Fungus. - The equipment and assemblies shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. For gasket or other non-hermetically sealed equipment, the parts and materials of the chasis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in this open condition.

4.9.6 Salt Fog. - The equipment and assemblies shall be exposed in their operating configuration to salt fog in accordance with Method 509 of MIL-STS-810 for 48 hours. Equipment and assemblies shall not be opened and shall not be required to be operated.

4.10 Inspection for interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) Impeller mates with shafts.

(b) Vertical support and connector housing mate properly.

(c) Tail section mates with center section.

(d) Mast sections mate properly.

(e) Size and form of special threads.

4.11 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine compliance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (See 6.1(b)).

5.1.1 Level A

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5.1.1.1 Cleaning. - Each Wind Measuring Set AN/AMQ-11 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each Wind Measuring Set AN/AMQ-11 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods specified in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Spare cap rubber, gasket, impeller assembly. - Each cap rubber gasket and impeller shall be individually packaged method III as follows: Cushion each item by wrapping in paperboard conforming to PPP-P-291, type 1, secure wrapping with gum kraft tape conforming to PPP-T-45, type III Grade A.

5.1.1.4.3 Consolidated package. - Items packaged as specified in 5.1.1.4.2 shall be placed within a corrugated fiberboard box conforming to PPP-B-636, W6c, weather resistant. Box closure shall be as specified in the appendix of the box specification.

5.1.1.4.4 Spare Fuses. - Fuses shall be packaged Method III as follows: Cushion the fuses by wrapping together in paperboard conforming to PPP-P-291, type 1, secure wrapping with gum kraft paper tape conforming to PPP-T-45, Grade A, Type III.

5.1.1.4.5 Spare pilot lamps. - Each lamp shall be individually packaged Method III as follows: Cushion each lamp by wrapping in paperboard conforming to PPP-P-291, type 1, secure the wrapping with gum kraft paper tape conforming to PPP-T-45, Grade A, Type III.

5.1.1.4.6 Dissembly from Transmitter. - Remove the impeller, hub and connector housing.

5.1.1.4.7 Impeller, Hub and Connector housing. - Each impeller, hub and connector housing shall be individually packaged method III as follows: Each item shall be cushioned on all surfaces with cells or pads or both, conforming to PPP-F-320 W5c, weather resistant, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W6c, weather resistant. Box closure-shall be as specified in the appendix of the box specification.

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5.1.1.4.8 Transmitter, Wind Direction and Speed Transmitter T-420( )/GMQ-11. - Each transmitter shall be packaged method III as specified in 5.1.1.4.7.

5.1.1.4.9 Consolidated Package. - Consolidate the items package in 5.1.1.4.3, 5.1.1.4.7 and 5.1.1.4.8 within a close-fitting fiberboard box conforming to PPP-B-636, M5c, weather resistant Box closure shall be as specified in the appendix of the box specification.

5.1.1.4.10 Indicator Wind Direction and Speed ID-373( )/GMQ-11. - Each indicator shall be packaged Method III as follows: Wind the cable into a coil of proportionate dimentions and tie in three places with tape conforming to T-T-871. Secure the coiled cable assembly and the spare parts packaged as specified in 5.1.1.4.4 and 5.1.1.4.5 to the indicator with tape conforming to PPP-T-76. Cushion the indicator on all surfaces with pads or cells or both conforming to PPP-F-320 designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close fitting fiberboard box conforming to PPP-B-636, W5c, weather resistant, box closure shall be as specified in the appendix of the box specification.

5.1.1.4.11 Disassembly of Support. - Disassemble the support by removing the upper mast, the lower mast, long braces and short brace, including the mounting feet. Hardware removed from this operation shall be placed together within a cloth draw-string bag conforming to PPP-B-20.

5.1.1.4.12 Components of Support. - Components shall be packaged Method III as follows: Form the masts and the braces into a neat bundle and secure with a minimum of 2 bands of steel strapping. The bag of hardware referenced in 5.1.1.4.11 shall be secured to the bundle by one of the steel bands. The steel strapping shall conform to QQ-S-781, Type 1, Grade 2, Class A.

5.1.2 Level C. - Wind Measuring Set AN/GMQ-11 shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B, or C as specified.

5.2.1 Level A

5.2.1.1 Wind Measuring Set AN/GMQ-11.

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5.2.1.2 Consolidation. - Transmitter and indicator shall be packed as follows: Place one each Transmitter, Wind Direction and Speed T-420( )/GMQ-11 packaged as specified in 5.1.1.4.9 and one each Indicator, Wind Direction and Speed ID-373( )/GMQ-11 packaged as specified in 5.1.1.4.10 together within a close-fitting box conforming to PPP-B-601, overseas type, PPP-B-621, Style 4 Class 2 or PPP-B-585, Style 2 or 3, Class 3. Place the technical literature packaged as specified in 5.1.1.4.1 on top of the contents, directly under the lid of the box closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, Type 1, Class B.

5.2.1.3 Support shall be packaged as follows: Place one each support package as specified in 5.1.1.4.12 and pack within a close-fitting nailed wood box conforming to PPP-B-621, Style 4, Class 2, closure and strapping shall be in accordance with the container specification or appendix thereto, except that metal strapping shall conform to QQ-S-781, Type 1, Class B.

5.2.1.4 When components of Wind Measuring Set are procured separately the following procedures shall apply:

A quantity of components bearing the same stock number packaged as specified in 5.1.1.4.7, 5.1.1.4.8, 5.1.1.4.9, 5.1.1.4.10 and 5.1.1.4.12 shall be packed in boxes as specified in 5.2.1.2 and 5.2.1.3.

5.2.2 Level B. - Wind Measuring Set AN/GMQ-11 packaged as specified in 5.1 shall be packed in accordance with 5.2.1.1.

5.2.3 Level C. -

5.2.3.1 Consolidation. - Wind Measuring Set AN/GMQ-11 packaged as specified in 5.1 shall be packed as specified in 5.2.1.1, except that the shipping containers shall be PPP-B-601, Domestic, PPP-B-621, Class 1, and PPP-B-585, Class 1, Box closure shall be in accordance with the applicable box specifications.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of preservation, packaging and marking required. (See Section 5).

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(c) Preproduction inspection:

(1) Four samples of each item cited in Section 1 are generally required so that preproduction testing can be performed in a reasonable amount of time. The following table is a recommended plan for the use of each sample.

Sample #1	Sample #2	Sample #3	Sample #4
Temperature Moisture-resistance Rain	Fungus	Complete performance Salt Fog	Complete performance To be retained as Standard

(2) The cost of samples which may be destroyed in tests will be borne by the contractor.

(3) When rough handling tests are required see Section 5.

(d) Marking and shipping of samples.

(e) Submission of the statement of treatment shall be submitted as soon as possible after award of contract to the contracting officer.

(f) Place of final inspection.

6.2 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example AN/GMQ-1W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.3 Dimensional data. - Sizes of packaging materials prescribed in Section 5 are based on the dimensions of the equipment cited on the applicable Bill of Material. When the dimensions of the equipment vary from those cited, the sizes of the packaging materials shall be adjusted accordingly. When shown in the Bill of Materials (See Section 5), corrugated fiberboard manufactured with A, B or C fluting may be used at the option of the contractor. When the fluting used is not the same as that cited in the Bill of Material, the dimensions of the affected packaging and packing materials will be adjusted accordingly.

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6.4 Definition of level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.5 Group C inspection. - Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2).

6.6 Verification inspection. - Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:

Army - EL

Preparing Activity

Army - EL

User Activities:

AF - 71

Project Number 6660-A019

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22 R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION
MIL-W-12600E(EL) WIND MEASURING SET AN/GMQ-11( ) (UNITS OF)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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